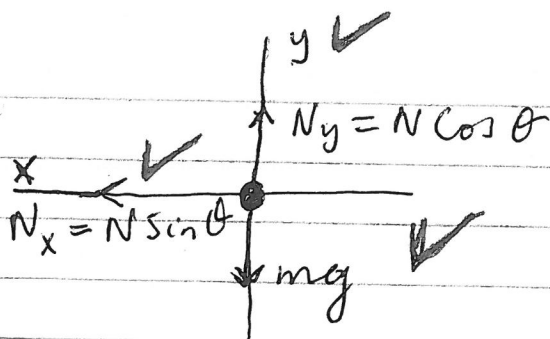
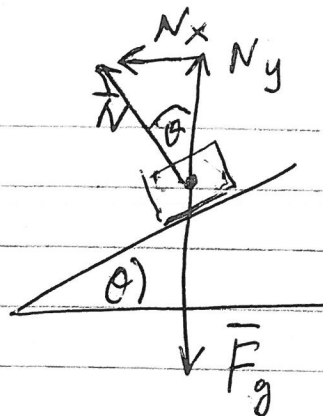


Q1



$$\sum F_r = N \sin \theta = \frac{mv^2}{r} \quad \checkmark \quad \text{--- (1)}$$

Newton's II applied to the radial direction, which is the x-direction.

$$\sum F_y = N \cos \theta - mg = 0 \quad \checkmark \quad \text{--- (2)}$$

Newton's II applied to the y-direction where we have equilibrium conditions.

$$\therefore \tan \theta = \frac{\frac{mv^2}{r}}{mg} = \frac{v^2}{rg} \quad \checkmark \quad \text{dividing (2) into (1)}$$

$$\therefore \theta = \tan^{-1} \left(\frac{v^2}{rg} \right)$$

$$= \tan^{-1} \left\{ \frac{\left[\frac{(60 \text{ km} \times 1000 \text{ m/km})}{1 \text{ hr} \times \frac{60 \text{ min}}{1 \text{ hr}} \times \frac{60 \text{ s}}{1 \text{ min}}} \right]^2}{(40.0 \text{ m} \times 9.80 \text{ m/s}^2)} \right\}$$

Subtract
1/2 for
incomplete
conversion

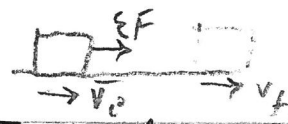
$$= 35.3^\circ \quad \checkmark$$

✓ 3 sig. fig.

9

NOTE: Students may use a tilted axes system. However, they need to recognise that 1/2 the circular path will be underground. And, multiplication by a trigonometric factor is necessary.

Q2



$$W_{\text{ext}} = \int_{x_i}^{x_f} \vec{F} \cdot d\vec{r} \quad \checkmark \quad \text{Take } d\vec{r} = dx \hat{i}$$

$$= \int_{x_i}^{x_f} F dx \quad \checkmark \quad \text{Since } \vec{A} \cdot \vec{B} = AB \cos \theta$$

$$= \int_{x_i}^{x_f} ma dx \quad \checkmark$$

$$= \int_{x_i}^{x_f} m \frac{dv}{dt} dx = \int_{x_i}^{x_f} m \frac{dv}{dx} \frac{dx}{dt} dx \quad \checkmark$$

You could give
them a mark
even if they don't
show

after re-expressing wrt velocity & changing
integration limits.

$$= \int_{v_i}^{v_f} m v dv \quad \checkmark$$

$$= m \left[\frac{1}{2} v^2 \right]_{v_i}^{v_f}$$

$$= \frac{1}{2} m v_f^2 - \frac{1}{2} m v_i^2 \quad \checkmark$$

$$= K_f - K_i = \Delta K.$$

2 parts for
good premises
or other
housekeeping.

(10)

Alternative approach

For simplicity, suppose the force acting on the block is constant $\vec{F} = F \hat{i}$ and the mass was initially at rest, at the origin.

The acceleration of the mass can be written as:

$$\vec{a} = \frac{\vec{F}}{m} = \frac{F}{m} \hat{i}$$

$$\text{The speed: } \vec{v}(t) = \int_{t=0}^t a dt = \frac{F}{m} t \hat{i} + 0$$

The block's position:

$$\vec{r}(t) = \int_{t=0}^t \vec{v}(t) dt = \frac{F}{2m} t^2 \hat{i} + 0 \quad \text{after integrating}$$

PAGE 2a

Now, calculating the work done to get the block to move at speed v .

The time taken to reach v : $t = \frac{m}{F} v$ ✓

⇒ Work done by the force F up to time t :

$$W_{\text{ext}} = \int \vec{F} \cdot d\vec{r} = \int \vec{F} \cdot \frac{F}{2m} (t')^2 \vec{i} dt' \quad \checkmark$$

Alternative
formulation -
(10)

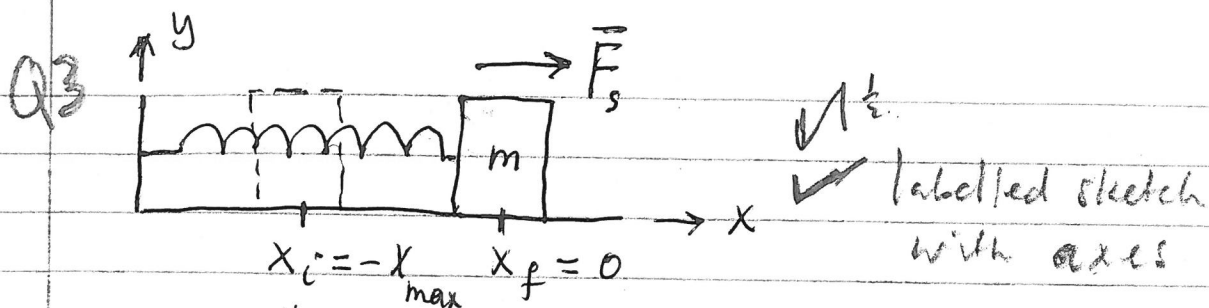
$$= \frac{F^2}{2m} (t')^2 \quad \checkmark \quad \text{Since } t = \frac{m}{F} v$$

$$= \frac{1}{2} m v^2$$

$$\text{Generalising: } W_{\text{ext}} = \frac{1}{2} m v_f^2 - \frac{1}{2} m v_i^2 = K_f - K_i \quad \checkmark$$
$$= \Delta K$$

MARKING

① If the premise is weak or not set altogether, subtract (1) or (2), respectively.



$$W = \int_{x_i}^{x_f} \vec{F} \cdot d\vec{r} \quad \checkmark \quad \text{Definition of work done}$$

$$= \int_{x_i}^{x_f} F_s \hat{i} \cdot dx \hat{i} \quad \checkmark$$

$$= \int_{-x_{\max}}^0 (-kx) dx \quad \checkmark$$

since $\hat{i} \cdot \hat{i} = 1$

and $\vec{A} \cdot \vec{B} = |\vec{A}| |\vec{B}| \cos \theta$

and $\theta = 0^\circ \Rightarrow \cos 0 = 1$

$$= (-k) \left[\frac{x^2}{2} \right]_{-x_{\max}}^0$$

$$= (-k) \left[0 - \frac{1}{2} (-x_{\max})^2 \right]$$

$$= \frac{1}{2} k x_{\max}^2 \quad \checkmark$$

(b) (i) The block and the spring $\checkmark$

$$(ii) \Delta E_{\text{system}} = \Delta K + \Delta U$$

$$= \left(\frac{1}{2} m v_f^2 - \frac{1}{2} m v_i^2 \right) + \left(\frac{1}{2} k x_f^2 - \frac{1}{2} k x_i^2 \right)$$

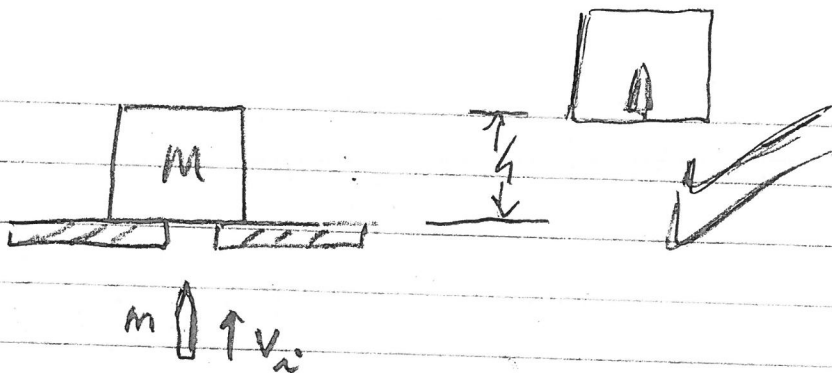
(c) System: Block

$$W_{\text{ext}} = \Delta K$$

$$\int_{x_i}^{x_f} \vec{F} \cdot d\vec{r} = \frac{1}{2} m v_f^2 - \frac{1}{2} m v_i^2 \quad \checkmark$$

9

Q4



Momentum before and after collision is conserved. However kinetic energy is not.

$$\vec{P}_{1,i} + \vec{P}_{2,i} = \vec{P}_{1,f} + \vec{P}_{2,f}$$

$$\Rightarrow m_1 v_{1,i} + m_2 v_{1,i} = m_1 v_{1,f} + m_2 v_{2,f}$$

$$\therefore m_1 v_{1,i} + M(0) = m v + M v = (m_1 + M) v$$

$$\Rightarrow v_{1,i} = \left(\frac{m_1 + M}{m_1} \right) v$$

$$\text{where } v = v_{1,f} = v_{2,f}$$

To find the initial velocity of the bullet, we need to find a relationship between v and the maximum height that the bullet-block system rises to. To do this, we use energy considerations.

Considering a system consisting of bullet, block and earth, this is an isolated system whose total energy is conserved.

$$\text{i.e. } \Delta E = 0 = \Delta K + \Delta U$$

$$\Rightarrow K_i + U_i = K_f + U_f$$

$$\frac{1}{2} (m + M) v^2 + 0 = (m + M) g h$$

$$\Rightarrow v = \sqrt{2gh}$$

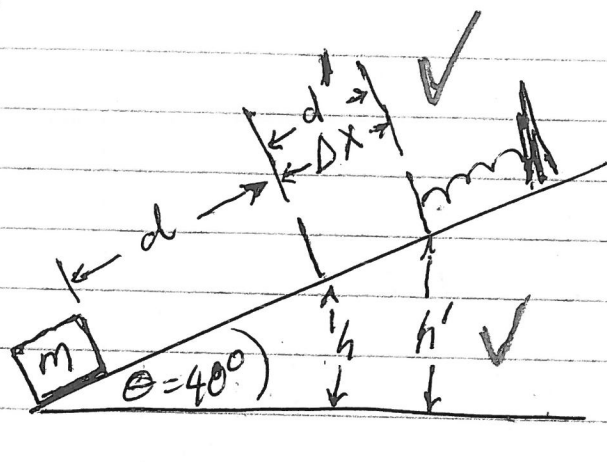
Some students may confuse the $1/2$ in the energy part of this problem.

$$\Rightarrow v_i = \frac{m+M}{m} \sqrt{2gh}$$

$$= \left(\frac{1.255 \text{ kg}}{0.00500 \text{ kg}} \right) \sqrt{2(9.80 \text{ m/s}^2)(0.220 \text{ m})}$$

$$v_i = 521 \text{ m/s} \checkmark$$

12



$$k = 200 \text{ N/m}$$

$$m = 1.0 \text{ kg}$$

$$K_i = 16 \text{ J}$$

$$\Delta x_s = 0.20 \text{ m}$$

$$U_g = 0$$

We choose a system consisting of the block, the spring and earth. For this system — isolated — the mechanical energy will be conserved; and the potential energy has a gravitational as well as a spring/elastic component.

When the spring is compressed 0.20 m

$$d' = d + \Delta x = 0.60 \text{ m} + 0.20 \text{ m} = 0.80 \text{ m}$$

$$\text{And, } h' = d' \sin \theta = 0.80 \text{ m} \sin 40^\circ = 0.51 \text{ m}$$

$$\begin{aligned} (a) \therefore \Delta E_{\text{mech}} &= 0 = \Delta K + \Delta U_g + \Delta U_s \\ K_i + U_{g,i} + U_{s,i} &= K_f + U_{g,f} + U_{s,f} \end{aligned} \quad \text{OR}$$

We set the potential energy U_g at the ground to be equal to 0.

$$\Rightarrow K_i + 0 + 0 = K_f + mgh' + \frac{1}{2} kx_f^2$$

$$K_f = K_i - mgh' - \frac{1}{2} kx_f^2$$

$$= 16 \text{ J} - 1.0 \text{ kg} (9.80 \text{ m/s}^2) (0.51 \text{ m}) - \frac{1}{2} (200 \text{ N/m}) (0.20 \text{ m})^2$$

$$= 7.0 \text{ J}$$

(b) We use the same system as in (a) and its mechanical energy is also conserved.

$$\Delta E_{\text{mech}} = 0 = \Delta K + \Delta U_g + \Delta U_s \quad \checkmark$$

$$\therefore \cancel{K_i = K_f} \quad 0 = (K_f - K_i) + (U_{g,f} - U_{g,i}) + (U_{s,f} - U_{s,i})$$

$$\therefore K_i = K_f + U_{g,f} + U_{s,f} \text{ with } U_{g,i} = 0 \text{ and } U_{s,i} = 0$$

When the spring is compressed

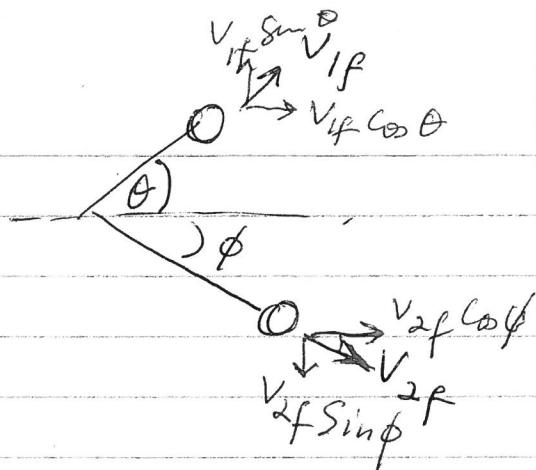
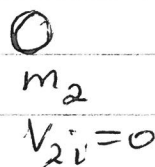
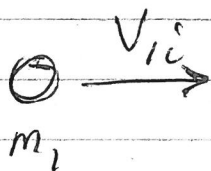
$0.60\text{ m} + 0.4\text{ m}$, the K_f of the block $= 0$

We need to determine the height h'' corresponding to this.

$$\therefore h'' = d'' \sin \theta = (0.60\text{ m} + 0.40\text{ m}) \sin 40^\circ = 0.64\text{ m} \quad \checkmark$$

$$\begin{aligned} \Rightarrow K_i &= 0 + mgh'' + \frac{1}{2} kx^2 \\ &= 1.0\text{ kg}(9.80\text{ m/s}^2)(0.64) + \frac{1}{2}(200\text{ N/m})(0.40\text{ m})^2 \\ &= 22.3\text{ J} \quad \checkmark \end{aligned}$$

(12)



(a) $\vec{P}_i = \vec{P}_f$ Conservation of momentum.

$$m_1 \vec{v}_{1i} + m_2 \vec{v}_{2i} = m_1 \vec{v}_{1f} + m_2 \vec{v}_{2f} \quad \checkmark$$

$$\vec{v}_{1i} + 0 = \vec{v}_{1f} + \vec{v}_{2f} \quad \text{since } m_1 = m_2$$

$$\frac{1}{2} m_1 v_{1i}^2 + \frac{1}{2} m_2 v_{2i}^2 = \frac{1}{2} m_1 v_{1f}^2 + \frac{1}{2} m_2 v_{2f}^2 \quad \checkmark$$

Conservation of Kinetic Energy
Since we have an elastic collision

$$\Rightarrow v_{1i}^2 + 0 = v_{1f}^2 + v_{2f}^2 \quad \checkmark \quad \text{since } m_1 = m_2 \quad (4)$$

(b) We will choose $\rightarrow x$ as $(+)$ and $\uparrow y$ as $(+)$.

$$x\text{-dim: } v_{1ix} \checkmark = v_{1fx} \cos \theta + v_{2fx} \cos \phi, \text{ and}$$

$$y\text{-dim: } 0 \checkmark = v_{1fy} \sin \theta + v_{2fy} \sin \phi$$

(The subscripts 'x' and 'y' are not necessary)

$$v_{2f} \cos \phi = v_{1i} - v_{1f} \cos \theta \quad \checkmark \quad (3)$$

$$v_{2f} \sin \phi = v_{1f} \sin \theta \quad \checkmark$$

(c) Squaring the equations and then adding them:

$$v_{2f}^2 \cos^2 \phi + v_{2f}^2 \sin^2 \phi = v_{1i}^2 - 2v_{1i}v_{1f} \cos \theta + v_{1f}^2 \cos^2 \theta + v_{1f}^2 \sin^2 \theta$$

$$\therefore v_{2f}^2 = v_{1i}^2 - 2v_{1i}v_{1f} \cos \theta + v_{1f}^2 \quad \checkmark$$

Substituting this into the Kinetic Energy conservation:

$$v_{1f}^2 + (v_{1i}^2 - 2v_{1i}v_{1f} \cos \theta + v_{1f}^2) = v_{1i}^2 \quad \checkmark$$

Patric G.

$$\therefore V_{if} (V_{if} - V_{ic} \cos \theta) = 0$$

(3)

$$\Rightarrow V_{if} = 0 \text{ or } V_{if} = V_{ic} \cos \theta \quad \checkmark$$

We discard the $V_{if} = 0$ solution because it doesn't correspond with our physical problem.

$$\therefore V_{if} = V_{ic} \cos \theta = (3.50 \times 10^5 \text{ m/s}) \cos 37.0^\circ \\ = 2.80 \times 10^5 \text{ m/s} \quad \checkmark \frac{1}{2}$$

$$\text{From } V_{ic}^2 = V_{if}^2 + V_{2f}^2$$

$$\Rightarrow V_{2f} = \sqrt{V_{ic}^2 - V_{if}^2}$$

$$= \sqrt{(3.50 \times 10^5 \text{ m/s})^2 - (2.80 \times 10^5 \text{ m/s})^2}$$

$$= 2.11 \times 10^5 \text{ m/s} \quad \checkmark \frac{1}{2}$$

$$\text{From } 0 = V_{if} \sin \theta - V_{2f} \sin \phi$$

$$\sin \phi = V_{if} \sin \theta / V_{2f}$$

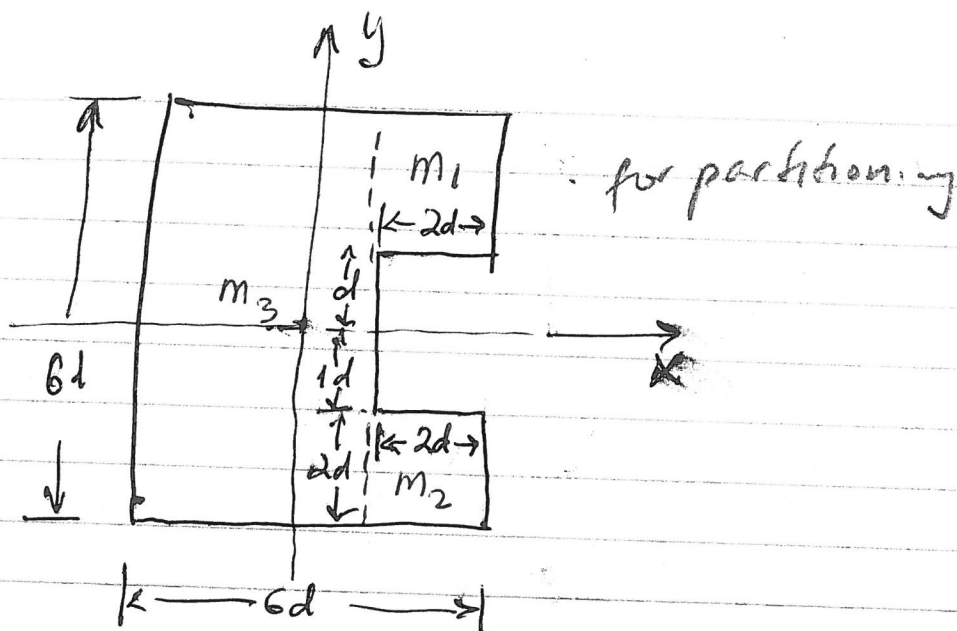
$$\phi = \sin^{-1} \left(\frac{V_{if} \sin \theta}{V_{2f}} \right)$$

(2)

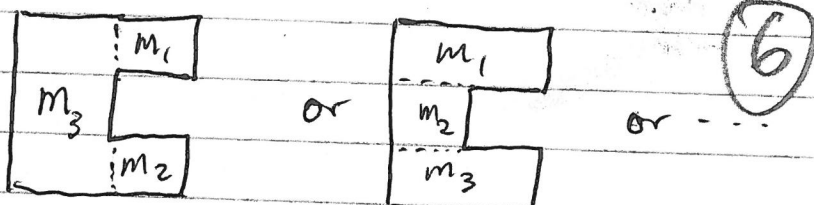
$$= \sin^{-1} \left(\frac{(2.80 \times 10^5 \text{ m/s}) \sin 37.0^\circ}{2.11 \times 10^5 \text{ m/s}} \right)$$

$$= 53.0^\circ \quad \checkmark$$

[12]
total



The plate could be divided into a number of convenient parts. For example



We need to know the ~~masses~~ masses of these pieces. Let each $2d \times 2d$ piece have mass m . Then, the $m_1 = m$; $m_2 = m$ and $m_3 = 6m$

Com of $m_1 = (2d, 2d)$ ✓
 Com of $m_2 = (2d, -2d)$ ✓
 Com of $m_3 = (-d, 0)$ ✓

The x_{cm} and y_{cm} are measured wr to the $x=0$ and $y=0$ reference lines respectively

$$\therefore X_{com} = \frac{1}{M} (m_1 x_1 + m_2 x_2 + m_3 x_3) \quad \text{from definition}$$

$$= \frac{1}{8m} [m(2d) + m(2d) + 6m(-d)] \quad r_{com} = \frac{\sum m_i r_i}{\sum m_i}$$

$$= \frac{1}{8m} (3dm) = \frac{3d}{8} \quad \checkmark$$

$$Y_{com} = \frac{1}{M} (m_1 y_1 + m_2 y_2 + m_3 y_3)$$

$$= \frac{1}{8m} [m(2d) + m(-2d) + 6m(0)] = 0 \quad \checkmark$$

PAGE 2